

Element F

Consideration of Design Viability

X-ray

The design is based on many factors that make it more functional. The doors are designed with a plastic slide and lock system to have the door so that the zoo can take X-rays from the side.

To perform X-rays from the top, the X-ray plates can be placed underneath the box. This allows the vet to easily change the position of the plate to change the picture of the X-ray. It also creates a simpler design for the overall box with fewer parts to wear out. For X-rays from the side, the X-ray plates need a way to be held on the side of the boxes. One design solution is to have a small ledge at the bottom of the box and support roughly 4 inches higher than the ledge on the box to keep the plates against the side of the box.

Ultrasound

The size and shape of the doors are created so that the vet will be able to easily perform ultrasounds on the dragons. The doors will have an opening system that can be X-rayed through to increase the function of the box. Through many design variations and ideas, prototypes were made. We agreed that a vertical sliding door was the best course of action considering all the criteria, including the x-ray, even with the increased complication of design. We continued to make a sliding system for the door. We decided to make 3d printed structures to hold the door to the boxes.

Storage

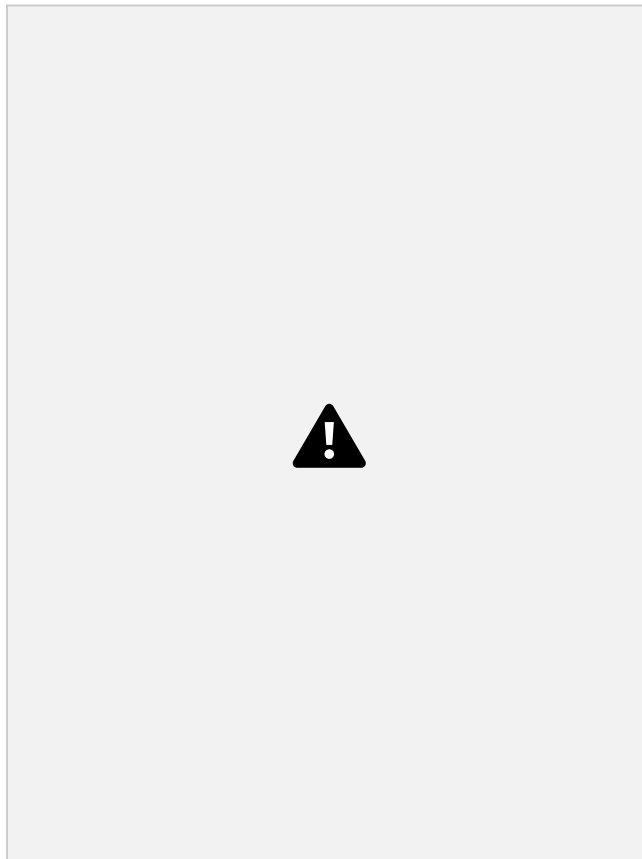
The current design is fitted to allow the storage of all sizes of dragons. With a modular design, the box will hold each size of dragon without giving them room to turn around or move their tail. These components allow for effective storage of dragons during transportation and examination of komodo dragons.

Materials

Plastic options are preferred over metal options in terms of design viability. When paired with examinations like X-ray and ultrasound, plastic options are best for not creating obstructions, also called artifacts, in the film -- a problem that would render the solution useless. [Ingraham CR et al.](#) describes the radiology of plastic: "The visibility of plastic is variable on both conventional radiography and computed tomography, depending on composition, but all types of plastic are well visualized on ultrasound." In contrast, the same article mentions the examination of aluminum: "Wood is most easily identified and localized on both computed tomography and ultrasound, is only faintly visible on conventional radiography, and is not well visualized on magnetic resonance imaging. Gravel, glass, and aluminum are well visualized on all modalities, with the exception of magnetic resonance imaging, where there is significant artifact surrounding the foreign

body.” This information provides clear evidence that plastic has proven to be an OP material for this project.

The guillotine door chosen for this project was one proven to be the best design of our viable options through the design matrices. In combination with latching, the guillotine door offers quick closing, durability, and reliability that other options lack. The following picture is a prototype of the guillotine door made by team members:



This design includes a plexiglass pane being slid into a 3D-printed PLA brackets that are attached to a larger plexiglass pane by nylon screws. In the picture, black metal screws are used as alternatives out of a lack of materials

and a need for a physical replacement; however, nylon screws will be used for all of the fixtures in the design for the final prototype. Three notches will be cut on each side in the shape of right triangles with the flat edge of the triangle perpendicular to the side of the plastic. The triangle will be reflected across the x-axis on either side of the plastic. These triangles will be filled with rubber triangles that will serve as stoppers for the plastic.

This design was reviewed by multiple expert opinions. Stacy Klein Gardner from Vanderbilt University agreed that the design would provide a safe and effective way to open and close the door. Jared Lawson of Vanderbilt University did research on what 3 objects look like under X-ray and he advised that we avoid layering plastic on top of

each other as it becomes easier to see plastic on x-ray the more thick it is. Bill Rodriguez from Vanderbilt University reviewed the design and assured members that the 3D printed brackets will withstand multiple uses and won't damage from repeated hits. He also suggested that we use a compliant mechanism for the latching detail.